



# Predictive Modeling of Root-Zone Variables Using Greenhouse Data

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## 01 Motivation

- Root-zone pH governs nutrient availability, while EC reflects salinity and fertilizer load - both directly affect crop health, growth and yield.
- In soilless greenhouse systems, the root zone has low buffering capacity, so irrigation or fertigation errors can quickly affect plant conditions.
- Manual laboratory pH/EC samples are accurate, but sparse, slow and reactive, leaving long blind gaps between measurements.
- Continuous probes can provide dense data, but they may drift, require frequent calibration and represent only a single point in the root zone.

## 02 Project Goal

- Build a machine-learning soft sensor that estimates future root-zone pH and EC between sparse manual laboratory samples.
- Predict the change from a known baseline measurement at  $t_0$  to a future target time  $t_1$  using climate, irrigation, fertigation, crop-state and previous pH/EC information.
- Provide reliable decision-support values that help growers detect pH or salinity drift earlier and maintain the root zone within an optimal operating range.

## 03 Data

16,682

10-min rows

109

pH / EC labels

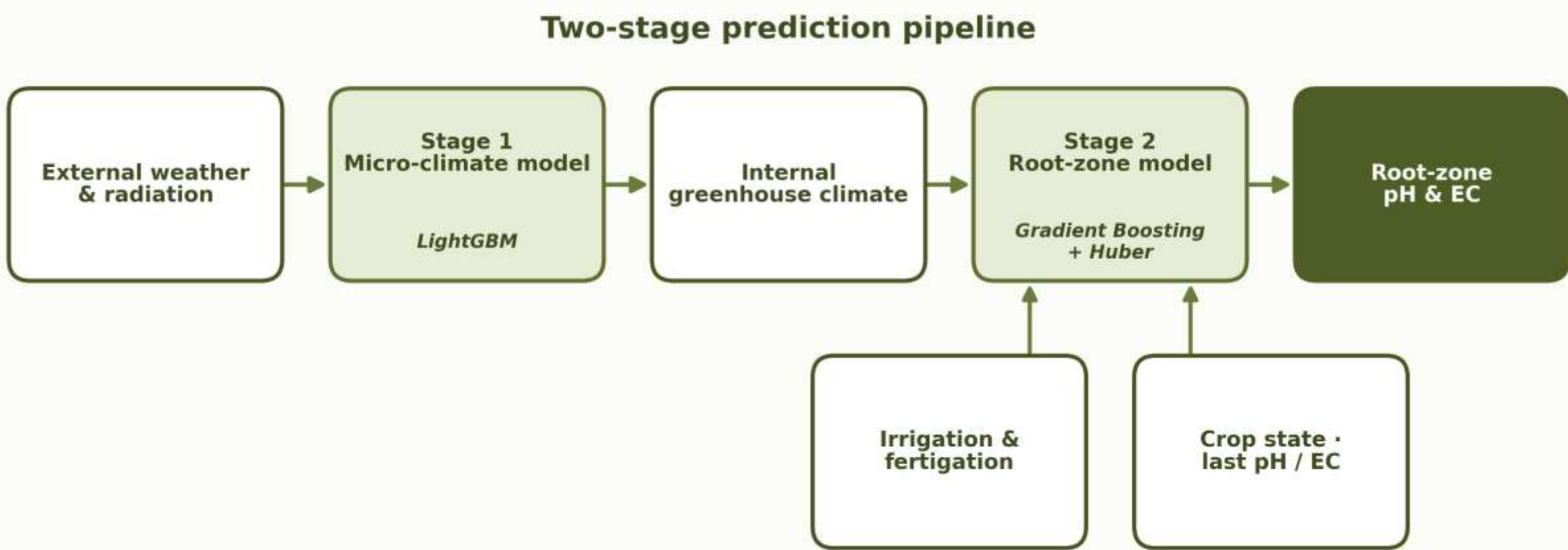
+20

input features

- Unified 10-minute master dataset · May 29 – Sep 21, 2025.
- External weather & solar radiation — open-field weather and radiation measurements.
- Greenhouse micro-climate — internal air temperature, relative humidity, radiation,  $ET_0$  and soil temperature.
- Irrigation & fertigation — water volume, fertilizer, EC, acid dosing and event timing.
- Crop-state features — canopy cover, crop development and time-history variables.
- Manual laboratory pH & EC measurements.

## 04 Method

- Two-stage soft sensor - Stage 1 predicts greenhouse micro-climate from external weather; Stage 2 predicts root-zone pH & EC.
- Models the  $t_0 \rightarrow t_1$  interval change rather than labelling every timestamp.



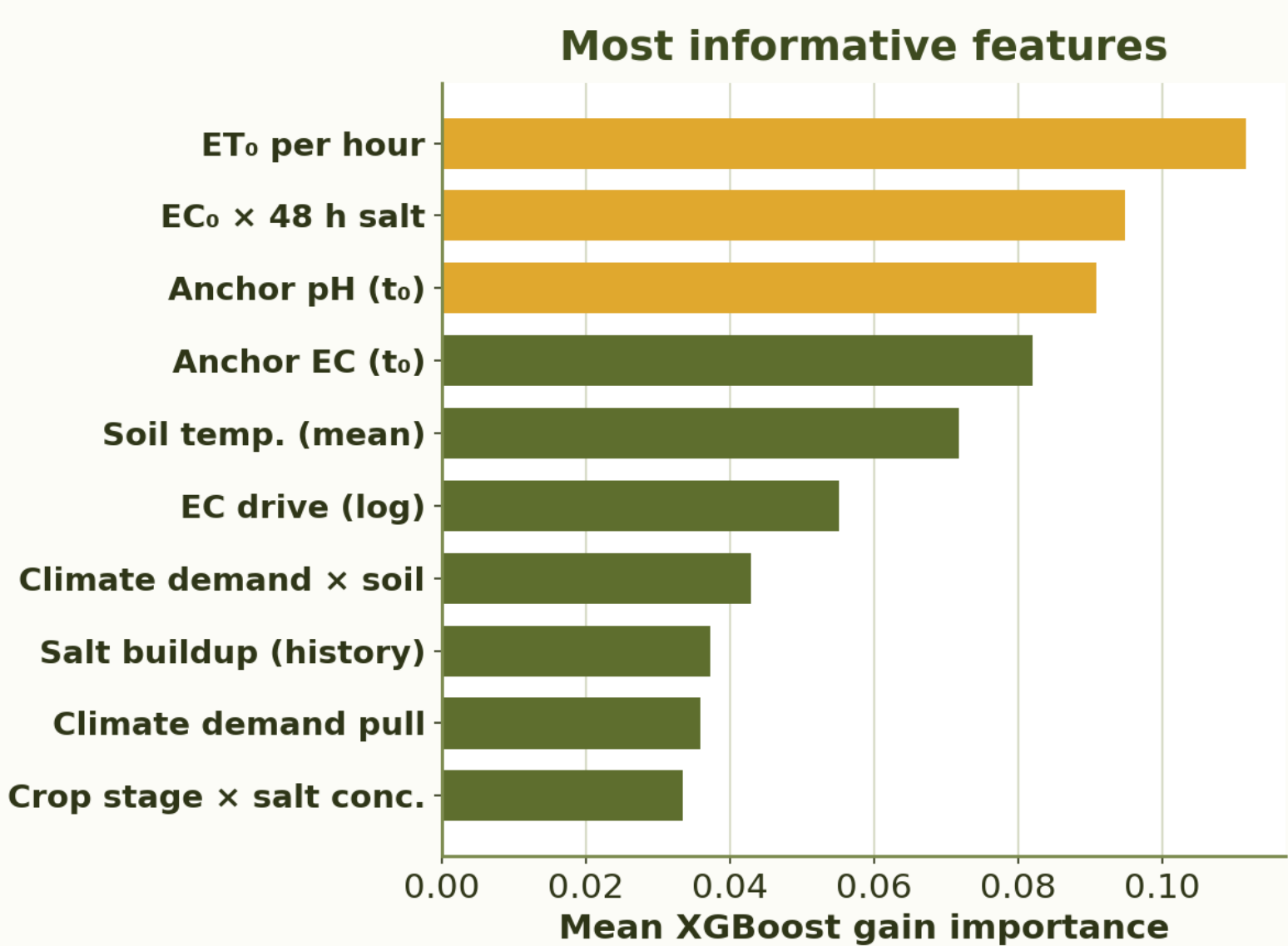
## 05 Algorithms & Model Choice

### Gradient Boosting

Performs well on tabular greenhouse data; captures nonlinear interactions between climate, irrigation, fertigation, crop state and previous pH/EC. Suited to the hybrid interpolation / extrapolation task.

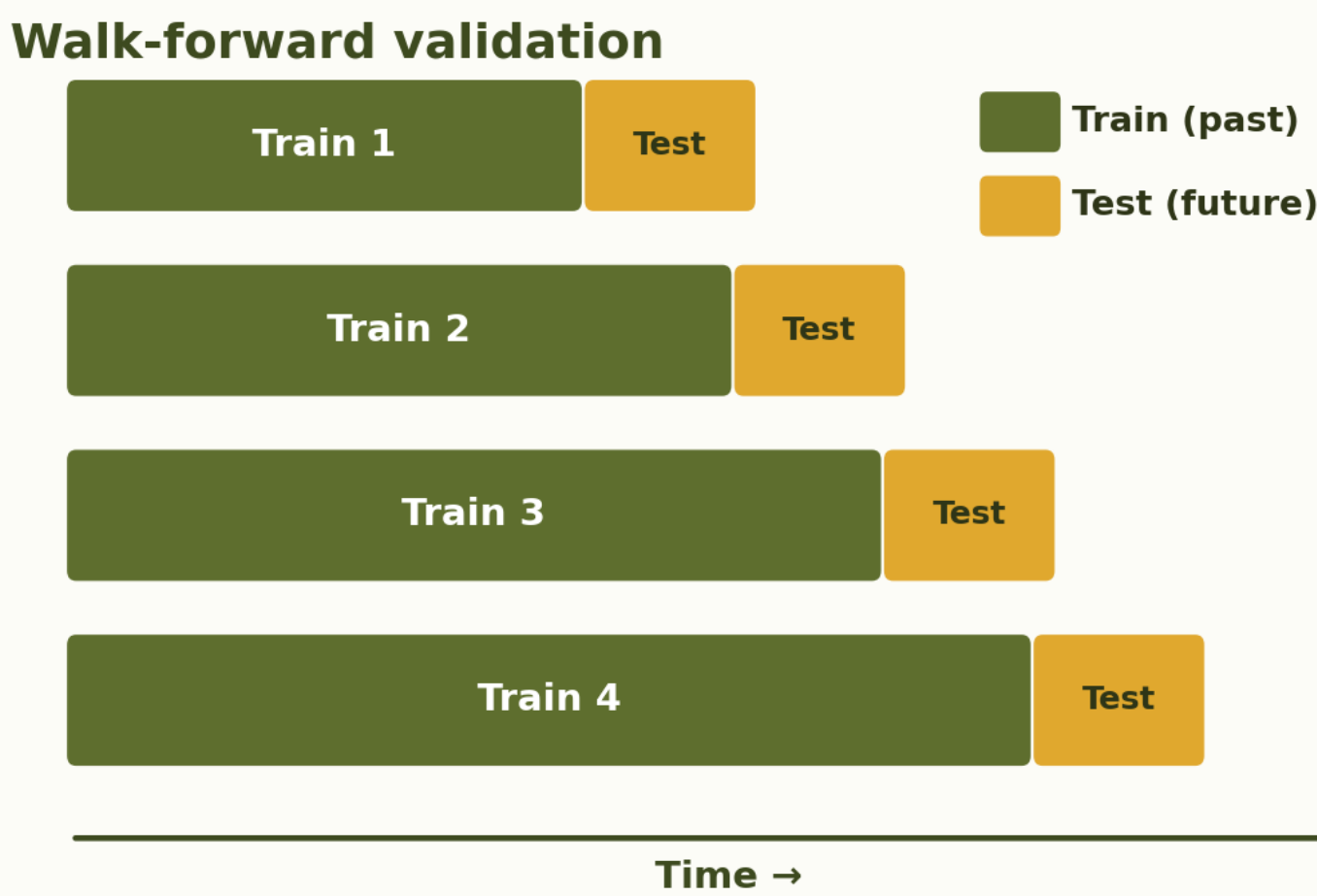
### Huber Regressor

Robust to outliers and the sharp changes caused by fertigation, acid injection or salinity shifts. A stable, complementary model that handles noisy and extreme target behavior.



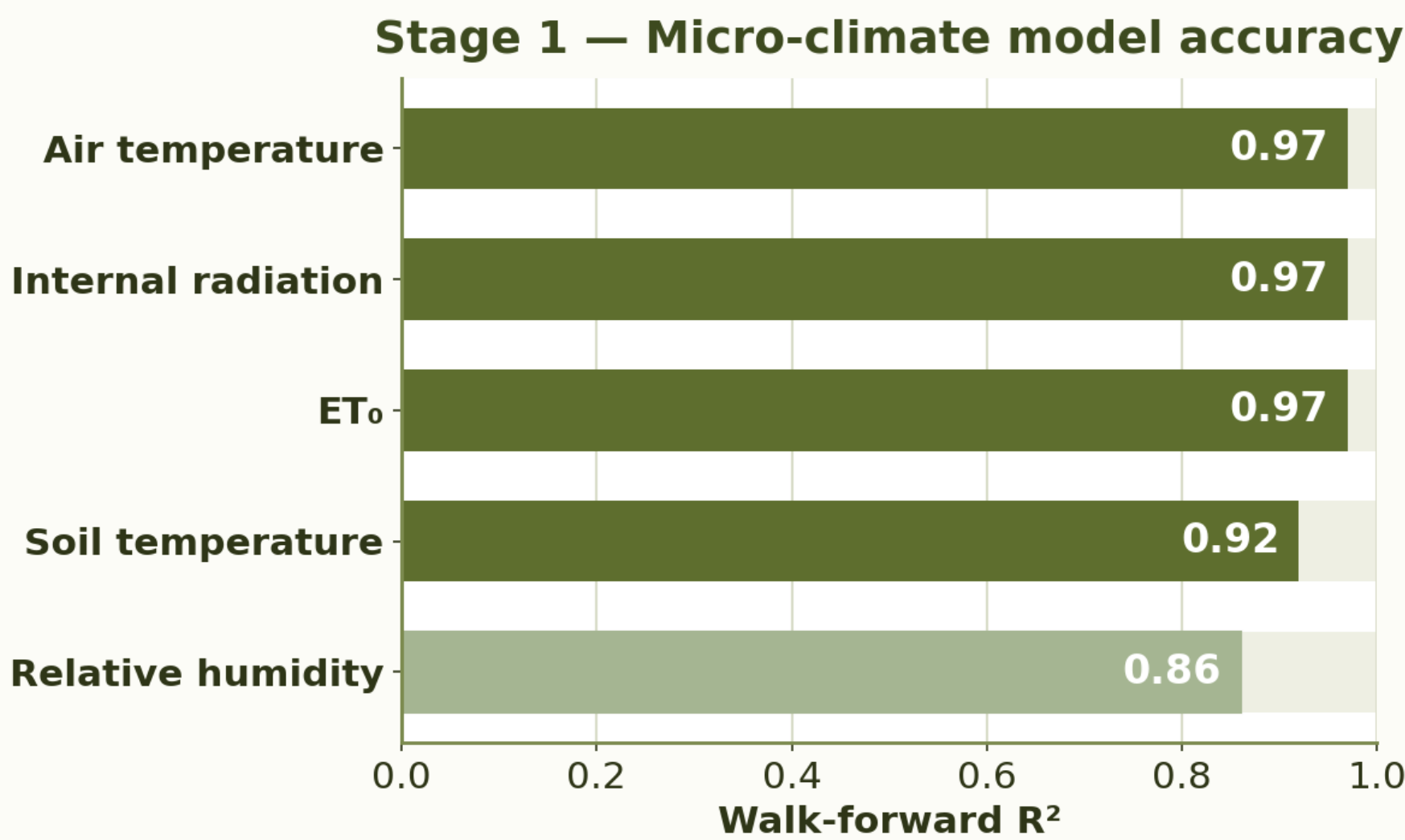
## 06 Validation Strategy

- Walk-forward validation: train only on past observations, predict future samples.
- At each step, the next test interval becomes part of the following training period.
- Metrics interpreted per target - pH (units) and EC (mS/cm).



## 07 Micro-Climate Model

- Predicts internal greenhouse climate from external weather and radiation.
- Generates continuous climate inputs - temperature, RH, radiation,  $ET_0$  and soil temperature - for the root-zone model.



## 08 Root-zone Model Results

0.33

pH MAE

0.93

pH  $R^2$

+62%

pH MAE gain

0.127

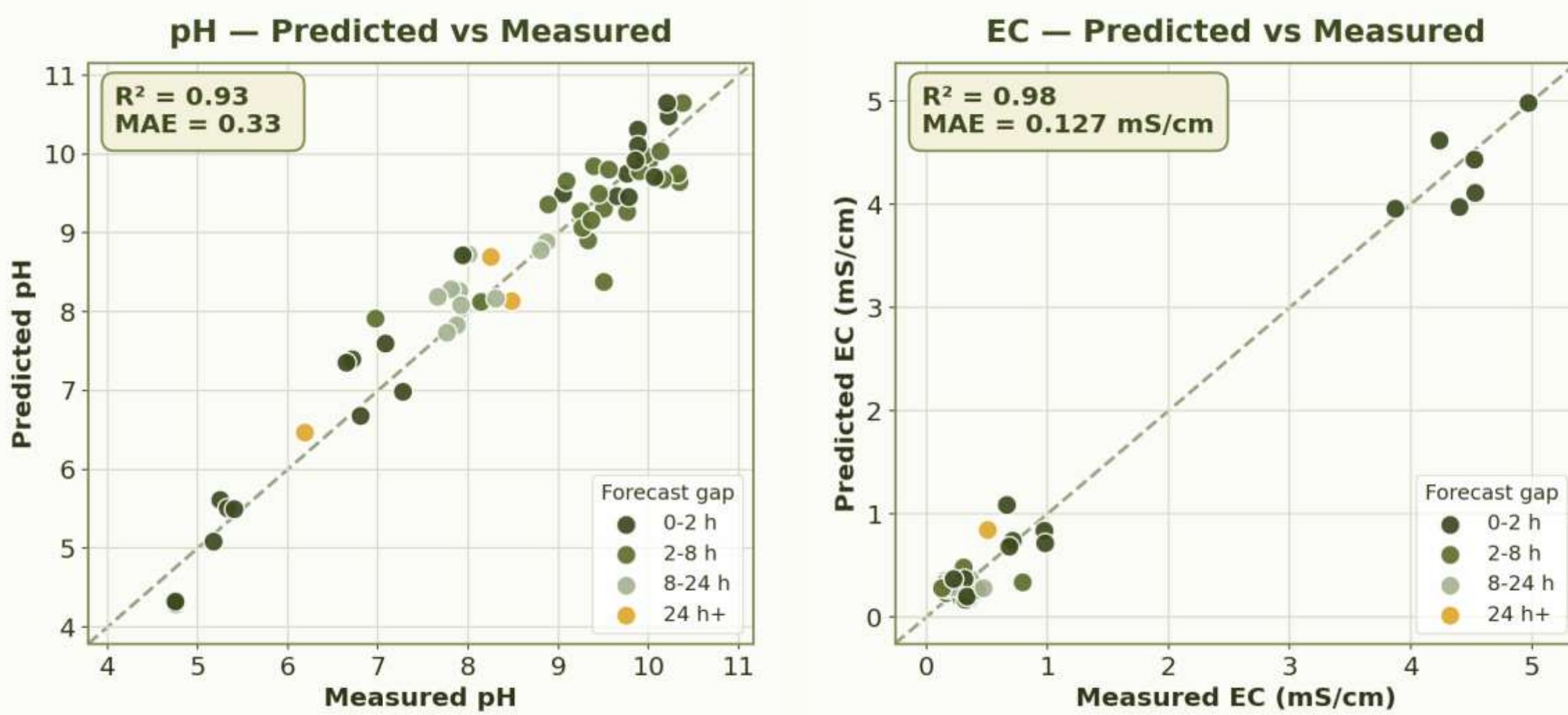
EC MAE

0.98

EC  $R^2$

+35%

EC MAE gain



## 09 Conclusions

- Acts as a soft sensor, estimating root-zone pH & EC between sparse manual lab samples.
- The hybrid two-stage approach improves accuracy over the last-measurement baseline (+62% pH, +35% EC).
- Designed for decision support - guiding irrigation & fertigation - not autonomous control.

## 10 Challenges & Limitations

- Only 109 manual pH/EC labels against 16,682 dense sensor records.
- No continuous ground-truth pH/EC between lab samples.
- Irregular sampling intervals complicate interval modelling.
- Higher uncertainty in rare or extreme fertigation / salinity cases.
- Limited generalization - data from a single season and one crop.
- More seasons, crops & greenhouse conditions needed for stronger validation.